

07_ReferenceImageSpec.md

MonitorAI v1.5 Overlay Guide Specification

Version: 1.5

Status: Baseline

1. Purpose

MonitorAI uses overlay guide images to improve capture consistency.

The overlays are not diagnostic tools.

The overlays are positioning aids only.

Purpose:

- Improve reproducibility
 - Improve visibility
 - Reduce retakes
 - Improve dataset quality
-

2. Overlay Philosophy

The overlay should help the patient understand:

Where to place the teeth.

Where to move the head.

How much anatomy should be visible.

The overlay should never be used for scoring.

Visibility scoring is performed separately by PTG.

3. Overlay Format

File Type:

PNG

Background:

Transparent

Orientation:

Landscape

Recommended Resolution:

2778 × 1284

Aspect Ratio:

iPhone Landscape

4. Overlay Opacity

Default:

50%

Allowed Range:

30%–70%

The patient must always be able to see their live image beneath the overlay.

5. Position Calibration Overlay

Displayed before Step01.

Purpose:

Align retractor opening with screen center.

Display Elements

3 × 3 Grid

Center Marker

Opening Detection Frame

Instruction

撮影口を画面中央に合わせてください

6. Overlay Asset List

The following overlays are required.

Group A

Aligner ON

5 mm Open

Step01

Front_ON.png

Step02

Right_Buccal_ON.png

Step03

Left_Buccal_ON.png

Group B

Aligner OFF

5 mm Open

Step04

Front_OFF.png

Step05

Right_Buccal_OFF.png

Step06

Left_Buccal_OFF.png

Step07

Upper_Occlusal.png

Step08

Lower_Occlusal.png

Group C

Occlusion

Light Bite

Step09

Front_Occlusion.png

Step10

Right_Occlusion.png

Step11

Left_Occlusion.png

7. Front Overlay Design

Used For:

- Step01
- Step04
- Step09

Purpose:

Guide frontal positioning.

Requirements:

- Midline visibility
 - Incisor visibility
 - Symmetric positioning
-

8. Buccal Overlay Design

Used For:

- Step02
- Step03
- Step05
- Step06
- Step10
- Step11

Purpose:

Guide posterior visibility.

Requirements:

- Posterior tooth visibility
 - Molar visibility
 - Arch continuity
-

Important

The smartphone should remain relatively stable.

The patient rotates their head.

The overlay should guide head rotation rather than camera movement.

9. Occlusal Overlay Design

Used For:

- Step07
- Step08

Purpose:

Guide arch visibility.

Requirements:

- Full arch visibility
 - Posterior arch visibility
 - Occlusal surface visibility
-

Important

Upper Occlusal:

Patient tilts head upward.

Lower Occlusal:

Patient tilts head downward.

The smartphone should remain relatively stable.

10. Overlay Color

Recommended:

Cyan

Purpose:

High visibility without obscuring anatomy.

Alternative

Yellow

Allowed for accessibility.

11. Overlay Positioning

Overlay images should be centered.

No automatic scaling based on PTG.

No distortion.

Maintain original aspect ratio.

12. Overlay Animation

Overlay images should remain static.

Avoid:

- Flashing
- Pulsing
- Excessive movement

Purpose:

Reduce distraction.

13. Overlay Interaction

The overlay does not receive touch input.

Touch interactions should be handled by UI controls only.

14. Overlay and PTG

Overlay:

Positioning Guide

PTG:

Visibility Evaluation

These systems must remain independent.

15. Future Expansion

Future versions may support:

- Adaptive Overlays
- Personalized Overlays
- Dynamic Visibility Guides

Version 1.5 uses fixed PNG overlays only.

16. Final Principle

Overlay guides exist to improve consistency.

The objective is not perfect visual alignment.

The objective is standardized clinical image acquisition suitable for future AI dataset generation.